

# Cybersecurity Incident Report:

## Network Traffic Analysis

Part 1: Provide a summary of the problem found in the DNS and ICMP traffic log.

The UDP protocol reveals that:

The browser sent out a DNS query to the DNS server (IP address 203.0.113.2) using the UDP protocol. This query was meant to retrieve the IP address for the website [www.yummyrecipesforme.com](http://www.yummyrecipesforme.com), as part of the standard process for resolving domain names to IP addresses. UDP is the transport protocol that DNS typically uses, and the packet was directed to port 53, which is reserved for DNS service.

This is based on the results of the network analysis, which show that the ICMP echo reply returned the error message:

The network analysis showed that instead of receiving a valid DNS response, the browser received an ICMP error message stating "udp port 53 unreachable." This error message indicates that the DNS request sent via UDP could not be delivered successfully because the target port (port 53) on the DNS server was not accepting the connection.

The port noted in the error message is used for:

Port 53, which is the standard port used by the Domain Name System (DNS) to process queries and respond to requests for resolving domain names into IP addresses.

The most likely issue is:

The DNS server at IP address 203.0.113.2 is either not running its DNS service, is misconfigured, or the service has crashed. As a result, no process is listening on port 53 to handle DNS queries, which is why the server is rejecting the request with an ICMP error message. This prevents the browser from resolving the website's domain name and loading the webpage. It's a DNS server failure.

## Part 2: Explain your analysis of the data and provide at least one cause of the incident.

Time incident occurred:

The incident occurred at 1:24 pm 32 seconds. based on the timestamp from the tcpdump log: 13:24:32.192571.

Explain how the IT team became aware of the incident:

After multiple reports of their website not working the IT team investigates using tcpdump. When using a packet sniffer or network analysis tool, you can see the logs for all incoming and outgoing traffic. Since port 53 couldn't be reached the error was sent via ICMP they know something is wrong.

Explain the actions taken by the IT department to investigate the incident:

To investigate, the IT department attempted to reproduce the problem by visiting the same website and encountered the same error. They used a network analysis tool (tcpdump) to capture the traffic between the client machine and the DNS server. They closely examined the outgoing DNS query packets and the responses from the DNS server to determine the root cause of the issue.

Note key findings of the IT department's investigation (i.e., details related to the port affected, DNS server, etc.):

**This means that the DNS service on the server was not responding to queries as expected, preventing the outgoing request for the website's domain name.**

Note a likely cause of the incident:

The likely cause is that the DNS service on the DNS server (203.0.113.2) was down, misconfigured, or had crashed. This could have been from a variety of things such as the different DOS or DDOS Attacks. We don't see any evidence of flooding traffic but the DNS server could have already been crashed for some time due to heavy traffic. There is no way to know for sure with the information given, you'd need evidence of massive spikes in traffic and logs showing thousands of requests.